Teacher guide

Timings, set-up, safety and classroom tips

Course at a glance

#	Lesson	Minutes	Needs a kit?
1	Welcome to Robotics 101	12	No
2	How do robots think? Sense, Think, Act	14	No
3	Explore your robot kit	15	No
4	How IR sensors detect a line	18	No
5	How motors move a robot	15	No
6	Build the robot body	16	Yes (or simulator)
7	Connect the electronics	18	Yes (or simulator)
8	Your first Arduino program	15	No
9	Read the IR sensors	15	No
10	Control the motors with code	15	No
11	The line-following algorithm	18	No
12	Make your robot follow the line	17	Yes (or simulator)
13	Fix common robot problems	15	No
14	Final challenge: show your robot	20	No
15	Final assessment and celebration	15	No

Total about 4 hours of guided time. Most schools run it as 8 to 10 class periods, plus building time.

Before the first class

- One kit per 2–3 students. Charge the batteries the day before.
- Install the Arduino IDE on lab computers and test an upload.
- Lay test tracks: black electrical tape (18–25 mm) on white chart paper, away from table edges and direct sunlight.
- Print the pin reference and troubleshooting guide for each table.

Safety rules

- Ask an adult before using batteries. Use only the battery pack from the kit.
- Never let the battery's red and black wires touch each other.
- Switch off and unplug power before changing any wire.
- Keep fingers, hair and loose clothes away from spinning wheels.
- If anything gets hot, smells or smokes: switch off and tell an adult.

Teacher checks

- ☐ Wiring checked before any battery goes in (Lesson 7 checklist).
- ☐ Wheels lifted off the table for first motor tests (Lesson 10).
- ☐ Batteries removed and stored at the end of each session.

Students without a kit

Every lesson works in the browser. The line simulator covers the experiments, and the final challenge has a virtual pathway: students finish the S-curve and sharp-turns tracks, and their results are recorded automatically as evidence.

Reviewing the final challenge

Projects appear under **Robotics & AI → Project reviews**. Approve, or send back with a note; students can resubmit. For the "teacher watched it run" pathway, approve only after you or a colleague have seen the run. Use the assessment rubric.

Classroom tips

- Pairs work best: one reads the steps, one builds; swap each lesson.
- Expect problems in Lesson 12. That is the lesson. Ask "what did you change?" before helping.
- Celebrate the first robot that finishes a lap, then the first that handles the sharp corner.